CHEMISTRY & SPECTROSCOPY SERIES

# Flame Test Colours

Unlocking Nature's Chemical Rainbow & Discovering the Magic of Colours in Fire

## 1. Introduction: Diwali Fireworks & Flame Magic

Have you ever wondered why fireworks sparkle in so many vibrant colours during Diwali and festive celebrations? From brilliant yellow bursts to vivid blue-green sparks and deep violet flashes, every light in the night sky is a chemical clue—a secret message emitted by specific metallic elements!

## 2. What is Spectroscopy?

### The Jump & Flash Dance of Atoms

**Spectroscopy** is the scientific study of how light and matter interact. When certain metal salts or compounds are exposed to a high-temperature flame, their atoms absorb thermal energy.

1. **Excitation:** The heat causes tiny electrons within the metal atoms to absorb energy and jump to higher energy levels (excited state).
2. **Emission:** These excited electrons are unstable and quickly fall back down to their original energy levels (ground state).
3. **Light Release:** As they fall, they release their extra energy in the form of light of a specific wavelength and color.

Because each element has a unique electronic arrangement, every metal performs its own distinct *"Jump & Flash Dance"*—producing a unique visual **colour fingerprint**!

## 3. Key Elements & Their Distinct Flame Colours

Scientists and analytical chemists rely on these characteristic color fingerprints to identify unknown elements in a sample during qualitative chemical analysis:

| Element | Chemical Symbol | Flame Colour | Real-World Everyday Sources |
|---|---|---|---|
| **Sodium** | $Na$ | Bright Yellow | Common table salt ($NaCl$), kitchen cooking salt, yellow street lights. |
| **Potassium** | $K$ | Lilac / Pale Violet | Festive fireworks (Sivakasi crackers), plant fertilizers, potassium salts. |
| **Copper** | $Cu$ | Blue-Green (Emerald) | |

| Element | Chemical Symbol | Flame Colour | Real-World Everyday Sources |
|---|---|---|---|
| | | | Copper water vessels, brass utensils, electrical wires, green fireworks. |
| **Calcium** | $Ca$ | Brick Red | Milk, bones, teeth, chalk, limestone, festive diyas and lamps. |
| **Strontium** | $Sr$ | Deep Crimson | Red emergency flares, red fireworks, specialized glass. |

**Kitchen Science Fact:** Next time you accidentally spill a pinch of common salt ($NaCl$) into your gas stove burner at home, you will instantly notice a bright yellow flame flicker! That is sodium displaying its signature electronic fingerprint right in your kitchen.

STUDENT ASSESSMENT SHEET

# Practice Questions: Spectroscopy & Flame Tests

Name: ______________________________ Date: ____________________

## Section A: Multiple Choice Questions

**Q1. Which element produces a characteristic bright yellow flame when heated?**

(a) Copper ($Cu$)

(b) Sodium ($Na$)

(c) Calcium ($Ca$)

(d) Potassium ($K$)

**Q2. What flame colour is observed when potassium salts are introduced into a flame?**

(a) Brick Red

(b) Blue-Green

(c) Lilac / Pale Violet

(d) Deep Crimson

**Q3. Copper compounds produce which distinct spark in a flame test?**

(a) Bright Yellow

(b) Blue-Green / Emerald

(c) Brick Red

(d) Pale Violet

**Q4. What causes the emission of light during a flame test?**

(a) Protons moving between different atomic nuclei

(b) Excited electrons returning to lower energy levels and releasing energy

(c) Chemical reactions creating new atomic nuclei

(d) Neutrons splitting inside the atom

## Section B: Matching & Short Answer Questions

**Q5. Match each element with its characteristic flame colour:**

| **Element:** | **Flame Colour:** |
|---|---|
| 1. Calcium ($Ca$) | A. Deep Crimson |
| 2. Strontium ($Sr$) | B. Lilac |
| 3. Sodium ($Na$) | C. Brick Red |
| 4. Potassium ($K$) | D. Bright Yellow |

Answers: 1 — _____, 2 — _____, 3 — _____, 4 — _____

**Q6. Explain why different metals emit different colours when placed in a burner flame.**

**Q7. An unknown salt sample from a fireworks factory is tested in a laboratory. It produces a brilliant blue-green flame. Identify the primary metal ion present in the sample and mention one everyday object made from this metal.**